

Predictive Modeling of Human Body Fat Percentage Using Regression Techniques

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Introduction

- Body Fat Percentage (BF%) is a key measure used in medical diagnostics, obesity analysis and athletic performance monitoring.
- Standard measurement techniques like hydrostatic weighing, BodPod, and DEXA are accurate but costly and inaccessible.
- Aim: Predict BF% using measurable anthropometric features through machine learning.
- Models used: Linear Regression and L2 Ridge Regression.

Dataset Description

- Source: Kaggle — Body Fat Prediction Dataset
- 252 samples, 14 predictor features, 1 target (BodyFat%)
- Very few people undergo scientific body composition testing
- Only **2%** of the global population exercises regularly in gyms and **less than 0.1%** undergo DEXA or hydrostatic body scans

Predictors	Density, Age, Weight, Height, Neck, Chest, Abdomen, Hip, Thigh, Knee, Ankle, Biceps, Forearm, Wrist
Target	BodyFat (%)

Train–Validation–Test Split and Standardization

- Train–Validation–Test Split: 70% / 20% / 10%
- Ensures unbiased generalization
- Scaling used: StandardScaler

$$Z = \frac{X - \mu}{\sigma}$$

- Standardization removes magnitude bias among features

Predictive Analysis Methods

- 1 Linear Regression — baseline regression model
- 2 Ridge (L2) Regression — prevents overfitting
- 3 Residual error analysis
- 4 Metrics: R^2 , RMSE

Histogram of Body Fat % Distribution

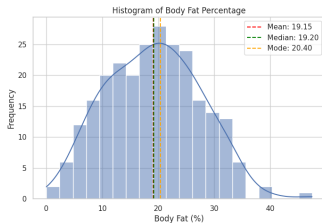


Figure: Body Fat% distribution showing mild right-skew

Statistical Summary

Mean = 19.15% Median = 18.20% Mode = 15%

- Skew handled through scaling to stabilize regression behavior

Linear Regression Model

$$\hat{y} = X\beta + b$$

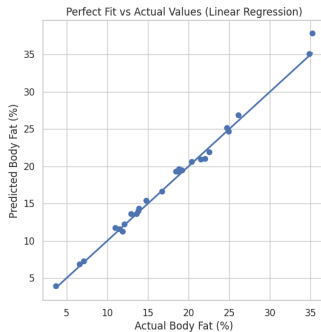


Figure: Actual vs Predicted Body Fat% — Linear Regression

L2 Ridge Regression

$$\hat{y} = X\beta + b, \quad \beta = \arg \min_{\beta} (||y - X\beta||^2 + \alpha ||\beta||^2)$$

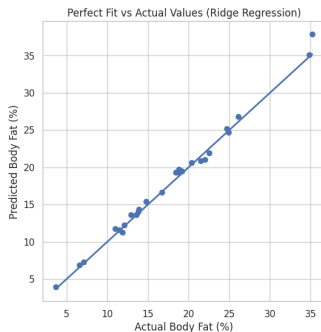


Figure: Actual vs Predicted Body Fat% — Ridge Regression

Why L2 (Ridge) over L1 (Lasso)?

- L1 performs feature elimination and becomes unstable with correlated variables
- L2 shrinks coefficient magnitude instead of eliminating features
- Handles multicollinearity common among circumference-based measurements
- Produces smoother and more generalized model weights

Residual Distribution Analysis

$$\text{Residual} = y_{\text{actual}} - y_{\text{predicted}}$$

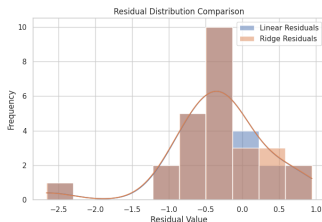
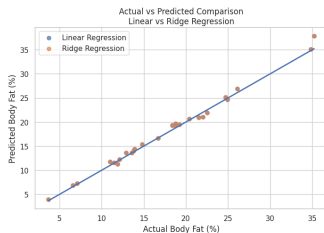


Figure: Residual Distribution — Linear vs Ridge Regression

- Ridge residuals centered around zero indicate superior model generalization

Linear vs Ridge Regression Comparison



Performance Comparison

Linear Regression $R^2 = 0.986$

Ridge Regression $R^2 = 0.992$

Body Density Calculation (Manual)

$$BD = 1.112 - 0.00043499(S7) + 0.00000055(S7)^2 - 0.00028826(Age)$$

- Jackson & Pollock 7-site skinfold (mm)
- Enables BF% prediction without laboratory scanning

Future Scope

- Models such as Random Forest, XGBoost, Neural Networks
- Dataset expansion across gender, age and ethnicity
- Wearable and mobile app integration
- Real-time fitness-tracking systems

Conclusion

- BF% can be accurately estimated using simple anthropometric measurements
- Ridge regression demonstrated better performance and stability
- Can be applied in fitness, sports science and preventive medicine

Thank You!

Questions?